
EECS 16A Designing Information Devices and Systems I

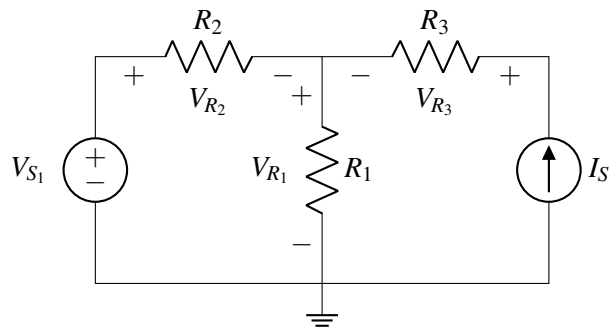
Discussion 9B

1. Superposition

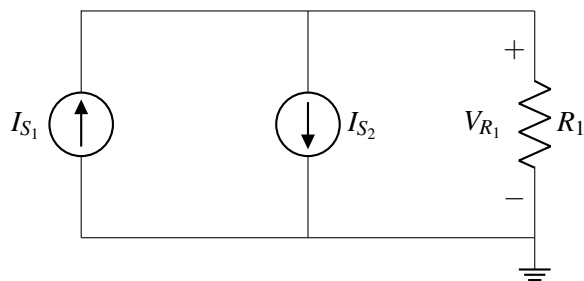
For the following circuits:

- Use the superposition theorem to solve for the voltages across the resistors.
- For parts (b) and (c) only, find the power dissipated/generated by all components. Is power conserved?

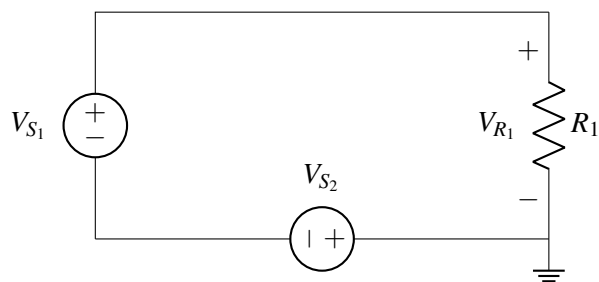
(a)



(b)

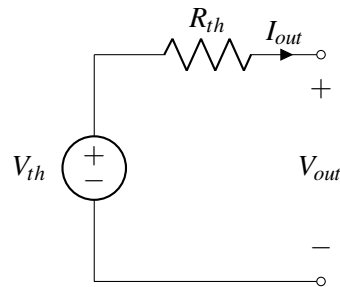
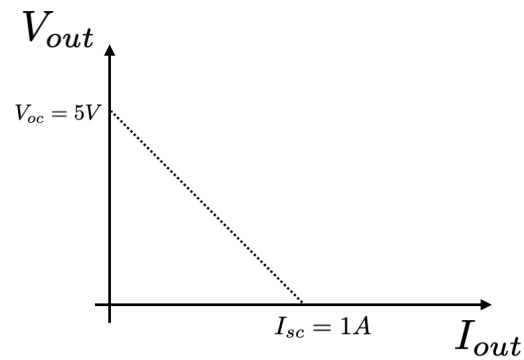


(c)



2. Thevenin equivalence

- (a) You are given the following $I_{out} - V_{out}$ characteristic of the Thevenin model of a circuit. Find the Thevenin voltage and the Thevenin resistance.



- (b) You are given a voltage divider as shown below. Find R_1 and R_2 such that the Thevenin equivalent model is the same as that of (a). You are given that $V_S = 10V$.

